

DHRUV VARSHNEY

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PROFESSIONAL SUMMARY

Detail-oriented Computer Science Engineering student with a strong foundation in full-stack development and computer vision. Proven innovator with a filed patent in gesture recognition technology and a track record of competitive problem-solving in hackathons. Aspiring to leverage expertise in Python, Java, and React to build scalable solutions at a forward-thinking tech organization.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript (ES6+), HTML5, CSS3

Frameworks/Libraries: OpenCV, NumPy, Pandas

Developer Tools: Git/GitHub, VS Code

Core CS Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOPs), Database Management Systems (DBMS)

KEY PROJECTS & INNOVATIONS

Hand Gesture Recognition System (Patented) Role: Lead Developer | Group Project |

[GitHub](#)

Sep 2025 - Nov 2025

Python, OpenCV, MediaPipe, NumPy, TensorFlow, PyAutoGUI

AI-powered Human-Computer Interaction (HCI) tool that enables touchless device control through real-time hand landmark tracking and gesture mapping.

- Engineered a real-time gesture recognition system using Python and OpenCV to facilitate touchless Human-Computer Interaction (HCI).
- Implemented MediaPipe for high-accuracy hand landmark detection, achieving seamless cursor control and system command execution.
- Patent Status: [Insert Patent Application Number] – Recognized for novel algorithmic approach in minimizing latency in low-light environments.
- Optimized the model to run efficiently on low-power hardware, reducing CPU overhead by 20%.

Gamified Education Platform – Smart India Hackathon | Role: Full-Stack Developer |

[GitHub](#)

Nov 2025 - Dec 2025

React.js, SQL, JavaScript (ES6+), CSS3

A rewards-based learning platform built to gamify the curriculum for primary education students.

- Architected the frontend using React.js, implementing a modular component structure that reduced initial page load time by 0.5 seconds. Designed a SQL database schema to handle 500+ concurrent user requests for real-time leaderboard updates and profile tracking. Developed a gamification engine that
- increased simulated student engagement scores by 40% through instant feedback quizzes and XP-based leveling systems.

CERTIFICATIONS & WORKSHOPS

- NPTEL Elite + Gold Certificate:** The Joy of Computing using Python | Score: 93% (Top 1% candidate). Jul-Oct 2025
- NPTEL Elite + Gold Certificate:** Programming in Java | Score: 94% (Top 1% candidate). Jan-Apr 2026
- Artificial Intelligence Workshop: Online | 2024**
 - Gained hands-on experience in Neural Networks, Model Training, and Data Preprocessing.

ACHIEVEMENTS & HOBBIES

- Achievements:** Qualified for SIH college level hackathon
- Interests:**
 - Chess:** Strategic player; focuses on pattern recognition and tactical planning.
 - Traveling:** Passionate about exploring diverse cultures and documenting experiences.

EDUCATION

B.Tech in Computer Science

GNIOT (Greater Noida Institute of Technology) | Greater Noida, UP | CGPA (First Year): 8.0

2024 - Present

Class XII (CBSE): Percentage: 88%

2024

Class X (CBSE): Percentage: 94%

2022